

# PG-CLNS — Pheromone-Guided Cooperative Large Neighborhood Search

Population of ants (ACS construction) coached by an adaptive LNS with simulated-annealing acceptance; the best member deposits pheromone that guides the next generation. Objective: maximise profit =  $R \cdot \bar{z}$  collected waste –  $C \cdot \bar{z}$  distance, subject to vehicle capacity and mandatory-bin coverage.

